AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102
Physics

Midterm Examination(MTA) – Makeup
Spring Semester 2024/2025

Date: 22/April/2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 1.5hrs

Instructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and** ID on the External Answer booklet.
- **Calculators** are allowed.

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PART I: Answer the entire Multiple Choice Questions (MCQ). [15 Marks]

1. The diameter of a circle is doubled. By what factor is the area changed?
a) √2 b) 2 c) 4 d) √4

2. Which of the following is not a base SI unit?
a) Kelvin b) kilogram c) newton d) second

3. What is the difference between velocity and speed?
a) Velocity is a scalar, speed is a vector
b) Speed is distance traveled / time, velocity is displacement / time
c) Velocity is always constant, speed can change
d) Velocity is directionless, Speed has direction

4. Dimensional analysis is used to:
a) Convert units b) Determine physical quantities
c) Analyze dimensions of physical equations d) All of the above

5. To be dimensionally consistent, force [ML/T^2], pressure [M/LT^2], and length [L] must be related as follows.
a) force = pressure^2 × length^2 b) force = pressure^2 × length
c) force = pressure × length^2 d) force = pressure × length

6. Which of the following is a vector quantity?
a) Mass b) Temperature c) Time d) Force

7. If an object undergoes constant acceleration, its velocity-time graph will be:
A) Parabola B) Straight line C) Curve D) Zigzag line

8. The total distance with respect to time is called:
a) Velocity b) Speed c) Acceleration d) Distance

9. What is the acceleration of an object if its velocity changes from 10 m/s to 30 m/s in 5 seconds?
a) 4 m/s² b) 6 m/s² c) 8 m/s² d) 10 m/s²

10. Which of the following is an example of free fall?
a) A ball rolling on a ramp b) A car moving on a straight road
c) A skydiver falling from an airplane d) A rocket launching into space

11. What happens to the speed of an object if the net force acting on it is zero?
a) Speed increases b) Speed decreases
c) Speed remains constant d) Speed fluctuates

12. In a collision between two objects, the total momentum:
a) Increases b) Decreases
c) Remains constant d) Depends on the masses of the objects

13. Newton's second law of motion describes the relationship between force, mass, and:
a) Displacement b) Velocity c) Momentum d) Acceleration

14. Impulse is defined as the:
a) Change in momentum b) Force applied in a period of time
c) Product of mass and acceleration d) Rate of change of velocity

15. The base SI unit of Momentum is:
a) Kg.m/Sec b) Kilogram c) Neuton.time d) m/Sec

16. Which of the following is an example of a contact force?
a) Gravitational b) Magnetic c) Frictional d) Electrostatic

17. The acceleration due to gravity on Earth is approximately:
a) 9.8 m/s^2 b) 3.14 m/s^2 c) 6.67 x 10^-11 m/s^2 d) 2.718 m/s^2

18. Which of the following is an example of a non-contact force?
a) Tension in a rope b) Normal force
c) Magnetic force d) Frictional force

ACCEPT

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19. The force required to accelerate an object of mass 6kg at a rate of 5 m/s² is
a) 1 N b) 20 N c) 30 N d) 5 N
20. Five children are pushing on a snowball; each pushes with a force of magnitude 10.0 N, and each is pushing at an angle of 45° relative to his neighbor. What is the magnitude of the net push on the ball?
a) 50 N b) 24.1 N c) 27.1 N d) 17 N
21. What happens to the momentum of an object if its mass doubles and its velocity remains constant?
a) Halves b) Doubles c) Quadruples d) No Change
22. Which of the following is an example of a balanced force?
a) Pushing a car uphill b) Tug of war between two teams
c) A book resting on a table d) Kicking a soccer ball
23. The net force acting on an object is zero. What can you conclude about its motion?
a) It is at rest b) It is moving with constant velocity
c) It is accelerating d) It is impossible to determine without more information
24. The force required to accelerate an object is directly proportional to its:
a) Mass b) Velocity c) Acceleration d) Displacement
25. If gravity between the sun and Earth suddenly vanished, Earth would continue moving in a/(an):
a) Curved path. b) Outward spiral path.
c) Straight-line path. d) Inward spiral path
26. The rate of change of velocity with respect to time is called:
a) Speed b) Acceleration c) Displacement d) Distance
27. If an object moves 8 meters east and then 5 meters west, the total displacement is:
a) 13 meters west b) 3 meters east c) 3 meters west d) 13 meters east
28. A car travels at 20 km/h for 3 hours. What distance does it cover?
A) 20 km B) 60 km C) 120 km D) 1800 km
29. An object is dropped from rest. What is its velocity after 2 seconds, neglecting air resistance?
A) 9.8 m/s B) 29.4 m/s C) 19.6 m/s D) 6 m/s
30. A rifle fires a bullet. Immediately after the bullet is fired, which of the following is not true?
a) The rifle and the bullet have the same magnitude of momentum.
b) The force on the rifle due to the bullet and the force on the bullet due to the rifle have the same magnitude.
c) The impulse on the rifle due to the bullet and the impulse on the bullet due to the rifle have the same magnitude.
d) The rifle and the bullet have the same kinetic energy.
31. Consider a car that travels between points A and B. The car's average speed can be greater than the magnitude of its average velocity, but the magnitude of its average velocity can never be greater than its average speed.
A) True B) False C) No enough data D) Depends on direction
32. A 10-kg rock and a 20-kg rock are dropped at the same time and experience no significant air resistance. If the 10-kg rock falls with acceleration a, what is the acceleration of the 20-kg rock?
A) 4a B) 2a C) a D) a/2 E) a/4
33. A runner ran the marathon (approximately 42.0 km) in 2 hours and 57 min. What was the average speed of the runner in m/s?
A) 14,200 m/s B) 124 m/s C) 3.95 m/s D) 14.2 m/s

AOU EXCEPT

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34. In a collision between a huge SUV and a small hybrid car, the SUV exerts a larger force on the hybrid than the hybrid exerts on the SUV.

A) True B) False

C) It depends on whether the collision is a head-on collision or a rear-end collision.

D) None of the above.

35. A 75-N box rests on a perfectly smooth horizontal surface. The minimum force needed to start the box moving is

A) 75 N. B) 7.5 N. C) 750 N. D) any horizontal force greater than zero.

36. A horizontal 52-N force is needed to slide a 50-kg box across a flat surface at a constant velocity of 3.5 m/s. What is the coefficient of kinetic friction between the box and the floor?

A) 0.11 B) 0.09 C) 0.10 D) 0.13

37. In a collision between two unequal masses, which mass receives a greater magnitude impulse?

A) the larger mass B) the smaller mass

C) They receive equal impulses D) None of the above

38. You are standing on a skateboard, initially at rest. A friend throws a very heavy ball towards you. You have two choices about what to do with the ball: either catch the ball or deflect it back toward your friend with the same speed as it was originally thrown. Which choice should you make in order to maximize your speed on the skateboard?

A) Catch the ball. B) Deflect the ball back.

C) Your final speed on the skateboard will be the same regardless whether you catch the ball or deflect the ball.

D) No enough information about speed

39. A rocket explodes into two fragments, one 25 times heavier than the other. The magnitude of the momentum change of the lighter fragment is

A) 25 times as great as the momentum change of the heavier fragment.

B) The same as the momentum change of the heavier fragment.

C) 1/25 as great as the momentum change of the heavier fragment.

D) 5 times as great as the momentum change of the heavier fragment.

40. Which of the following quantities are units of impulse? (There could be more than one correct choice.)

A) $N \cdot m$ B) $kg \cdot s/m$ C) $kg \cdot m/s$ D) $N \cdot s$ E) $kg \cdot m^2/s^2$

41. Identical forces act for the same length of time on two different objects. The magnitude of the change in momentum of the lighter object is

A) smaller than the magnitude of the change in momentum of the larger mass, but not zero.

B) larger than the magnitude of the change in momentum of the larger mass.

C) exactly equal to the magnitude of the change in momentum of the larger mass.

D) zero.

42. A 5-kg ball collides in-elastically head-on with a 10-kg ball, which is initially stationary. Which of the following statements is true? (There could be more than one correct choice.)

A) The magnitude of the change of velocity the 5-kg ball experiences is greater than that of the 10-kg ball.

B) The magnitude of the change of velocity the 5-kg ball experiences is less than that of the 10-kg ball.

C) The magnitude of the change of velocity the 5-kg ball experiences is equal to that of the 10-kg ball.

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D) The magnitude of the change of the momentum of the 5-kg ball is equal to the magnitude of

43. A 100-g ball falls from a window that is 12 m above ground level and experiences no significant air resistance as it falls. What is its momentum when it strikes the ground?

A) 3.3 kg · m/s B) 4.8 kg · m/s C) 1.8 kg · m/s D) 1.5 kg · m/s

44. A 60-kg swimmer suddenly dives horizontally from a 150-kg raft with a speed of 1.5 m/s. The raft is initially at rest. What is the speed of the raft immediately after the diver jumps if the water has negligible effect on the raft?

A) 0.60 m/s B) 0.15 m/s C) 1.5 m/s D) 3 m/s

45. In a closed system, Momentum is a conserved quantity?

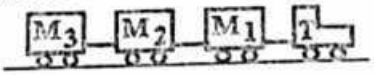
A) True B) True only for elastic collisions

C) False D) It depends on type of collided material

Part II: Answer the following questions. [15 Marks]

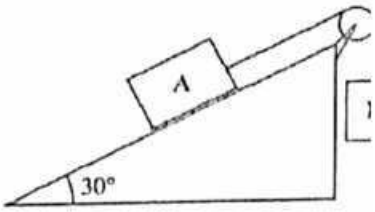
Question 1: (5 Marks)

In the figure, an airport luggage carrying train with a tractor T is pulling three luggage carts, M_1, M_2, and M_3 with constant velocity of 4.5 m/s. If T = 300 kg, M_1 = 200 kg, M_2 = 100 kg, and M_3 = 100 kg (there is no friction), then the force in the connection between the tractor T and cart M_1 is?



Question 2: (5 marks)

Two packages are connected by a very light string that goes over an ideal pulley as shown in the figure. Package A has a mass of 3.0 kg and can slide along a rough plane inclined at 30° above the horizontal. The string acts on package A parallel to the surface of the plane. The coefficient of static friction between package A and the plane is 0.40. What minimum mass should package B have in order to start package A sliding up the ramp?



Question3 (5 marks)

Two small steel balls A and B have mass 0.6 kg and 0.2 kg respectively. They are moving towards each other in opposite directions on a smooth horizontal table when they collide directly. Immediately before the collision, the speed of A is 8 m/s and the speed of B is 2 m/s. Immediately after the collision, the direction of motion of A is unchanged and the speed of B is twice the speed of A. Find;

(a) the speed of A immediately after the collision,

(b) the magnitude of the impulse exerted on B in the collision.

End of questions

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